

Premier University
Department of Computer Science & Engineering
B.Sc. 8th Semester, Midterm Examination, Spring 2025
Course Code: CSE 481 Course Title: CCCS, COs: CLO2, CLO5

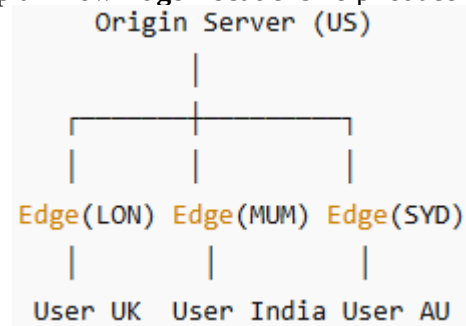
Time: 45 mins.

Answer any two sets of questions from the following three sets.

Total Marks: 20

- 1(a) Explain the difference between **AWS Regions and Availability Zones**. Use a simple diagram to illustrate how multiple AZs within a Region provide high availability. CLO2 05
- 1(b) In the context of the **AWS Shared Responsibility Model**, identify which of the following responsibilities belong to AWS and which to the customer. Justify your answer: CLO5 05
- Applying security patches to an EC2 instance
 - Physical security of AWS data centers
 - Configuring IAM policies for users
 - Oracle upgrades if the Oracle instance runs as an Amazon RDS instance
 - Securing the SSH keys to remote access through CLI or SDK.

- 2(a) Consider the diagram below showing an **AWS CloudFront distribution**. What is **CloudFront** and **Edge Location**? Explain how **Edge Locations** help reduce latency for global users. CLO2 05



- 2(b) A company wants to grant **temporary access** for its EC2 instance to read/write objects in an S3 bucket without embedding permanent credentials. CLO5 05
- Which IAM component should be used (User, Group, or Role)?
 - Briefly explain why it is the most secure choice.
- 3(a) AWS organizes its services into categories such as Compute, Storage, and Database. Choose one service from each category and explain how these three can work together to support your university's Academic Information Management Application (PUAIS). Draw a block diagram showing the interaction among Compute, Storage, and Database services. CLO2 05
- 3(b) Setting up a new AWS account requires several security best practices. Describe the steps you would follow to secure a newly created AWS account. Your answer should include: CLO5 05
- Proper use of the root account
 - Creation and management of IAM users and groups
 - Use of Multi-Factor Authentication (MFA)
 - Monitoring and logging with CloudTrail
- Why is it considered dangerous to use the root account for day-to-day operations? Provide justification.